

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)			microwaves	1			1		
	(b)			Indicative content: The Steady State Theory suggests that the Universe is always expanding with matter being continuously created to form new stars and galaxies to fill in the gaps Evidence The wavelength of the early radiation in the form of short wavelength/high energy radiation (gamma rays) has become longer wavelength / stretched (microwave) radiation. This is CMBR that presently pervades the Universe. Cosmological redshift means that light stretches as space expands. In fact, it stretches so much that by the time light gets to us from distant galaxies, absorption spectra from them show that black lines have moved to the red end of the spectrum. Both of the above support the Big Bang theory.	6			6		

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					AO1	AO2	AO3	Total	Maths	Prac
				5 – 6 marks Full description and how evidence disagrees with Steady State theory. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3 – 4 mark Reference to Big Bang and full description of CMBR or cosmological red shift or steady or partial description of more than one. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1-2 marks Any reference to Big Bang or cosmological red shift or steady state <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
				Question 7 total	7			7		